Frontend refactoring — IcfesApp (Flutter)

Branch	`refactor` (created from `app_test`, commit `817ad5b`)
Last update	2026-10-30
Backend	Must run the `refactor` branch of `quizAppIcfesBack` — see that repo's `docs/REFACTOR.md`

This document records every change made to the Flutter app during the refactoring: what changed, why, and how it works now.

> **Deploy together:** this frontend only works with the backend from its `refactor` branch, and the old frontend does not work with that backend (several endpoints were renamed or removed).
>
> **Language convention:** code, comments and docs in **English**; texts shown to the user stay in **Spanish**.

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1. Where the API is called

All HTTP calls go through `lib/core/network/api_client.dart` and three repositories:

Repository	File	Used by
`AuthRepository`	`lib/core/auth/auth_repository.dart`	`AuthController` (login, sign-up, session, password, email confirmation)
`ContentRepository`	`lib/features/content/data/content_repository.dart`	Admin panel, plans and subscription pages
`PracticeRepository`	`lib/features/practice/data/practice_repository.dart`	Student dashboard, practice session page

`ApiClient` adds the access token, and when a request answers 401 it renews the session once with the refresh token and retries.

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2. Change log

2.1 Endpoints updated to the new backend contract

Repository method	Before	Now
`AuthRepository.register`	`POST /api/auth/register`	`POST /api/auth/register-student`
`AuthRepository.session`	`GET /api/auth/session`	**Removed** — the user comes in the refresh-token response
`ApiClient` refresh	`POST /api/auth/refresh`	`POST /api/auth/refresh-token` (returns tokens **and user**)
`AuthRepository.logout`	`POST /api/auth/revoke`	`POST /api/auth/logout`
`AuthRepository.resendConfirmation`	`POST /api/auth/resend-confirmation`	`POST /api/auth/resend-confirmation-email`
`PracticeRepository.getMySessions`	`GET /api/practice-sessions` (full sessions)	`getHistory` → `GET /api/practice-sessions/history` (summaries)
`ContentRepository.revokeInvitationCode`	`POST /api/invitation-codes/{id}/revoke`	`PUT /api/invitation-codes/{id}` with `isActive: false`
`ContentRepository.getMySubscription`	`GET /api/subscriptions/current`	`GET /api/subscriptions/current-user`
`ContentRepository.getStudentReports`	`GET /api/reports/students`	`GET /api/reports/student-performance`

Routes that did not change: `login`, `register-staff`, `forgot-password`, `reset-password`, `change-password`, `confirm-email`, `/api/subjects`, `/api/institutions`, `/api/questions`, `/api/invitation-codes` (list/create/update/delete), `/api/plans`, `POST /api/practice-sessions`, `/answers` and `/finish`.

2.2 Session restore: 2 requests → 1

- **Before:** on startup `restoreSession` called `/refresh` (tokens) and then `/session` (user).
- **Now:** `/refresh-token` returns the same shape as login (tokens + user), so `ApiClient.tryRefreshSession()` returns the `AuthUser` directly (`null` if the session could not be renewed) and `restoreSession` is a single request.
- **Files:** `lib/core/network/api_client.dart`, `lib/core/auth/auth_repository.dart`.

2.3 Admin panel: update only what changed (no more full reloads)

- **Before:** after every create / update / delete, `_runMutation` called `_loadData()`, which re-downloaded the 5 lists (subjects, questions — with every option —, institutions, invitation codes and the report). Editing one subject name meant 6 requests.
- **Now:** the backend returns the created / updated object, and the panel changes only that item in its list:

Action	Requests now	What is updated locally
Create / update / delete question	1	The question in `_questions`
Create / delete subject	1	The subject in `_subjects` (kept sorted by name, like the API)
Update subject	1 (+1 if the name changed)	The subject; questions are reloaded **only if the name changed**, because they show `subjectName`
Create / delete institution	1	The institution; on delete its invitation codes are removed too (the backend deletes them in cascade)
Update institution	1 (+1 if the name changed)	The institution; codes are reloaded **only if the name changed**, because they show `institutionName`
Create / revoke / delete invitation code	1	The code in `_invitationCodes`

- Helpers in `admin_dashboard_page.dart` : `_replaceById` , `_removeById` , `_sortedSubjects` , `_sortedSchools` , `_updateIfMounted` .
- Repository `update*` methods now return the updated model (`SubjectModel` , `SchoolModel` , `QuestionModel` , `InvitationCodeModel`).
- The panel still loads everything once when it opens.

2.4 Question images are kept when editing 🐞

- **Risk found:** the backend now stores exactly the images it receives on update. `QuestionDraft` did not send any, so editing a question from the admin panel would have **deleted its images**.
- **Now:**
- `QuestionModel` / `QuestionOptionModel` read `imageUrl` and `justificationImageUrl` .
- `QuestionDraft` / `QuestionOptionDraft` send them back.
- The question dialog keeps each option's image in `_optionImageUrls` , a list parallel to the option fields, so adding or removing options never moves an image to a different option.
- **Files:** `lib/features/content/data/content_models.dart` , `lib/features/admin/presentation/widgets/admin_dialogs.dart` .

2.5 Lighter practice history

- **Before:** the student dashboard downloaded every session with all its questions and options, only to show "answered / total", status, subject and date.
- **Now:** `PracticeSessionSummaryModel` (`totalQuestions` , `answeredQuestions` , `correctAnswers` , status, dates) from `GET /api/practice-sessions/history` . The dashboard and the progress view use those counts.
- `PracticeSessionModel` (with questions) is still used by the practice session page, which gets it from `POST /api/practice-sessions` .
- **Files:** `lib/features/practice/data/practice_models.dart` , `practice_repository.dart` , `lib/features/student/presentation/pages/student_dashboard_page.dart` .

2.6 `School` → `Institution` (follows backend phase 6)

- **Why:** the backend renamed the concept everywhere and the user JSON field changed from `schoolId` to **`institutionId`**. Without this change `AuthUser.institutionId` would always be null.
- **API-visible change:** `AuthUser.schoolId` → `AuthUser.institutionId`, read from `json['institutionId']`.
- **Code renamed for consistency** (same concept, same name in back and front):

Before	Now
`SchoolModel`, `SchoolDraft`	`InstitutionModel`, `InstitutionDraft`
`ContentRepository.getSchools / createSchool / updateSchool / deleteSchool`	`getInstitutions / createInstitution / updateInstitution / deleteInstitution`
Admin panel: `_schools`, `_openSchool`, `_deleteSchool`, `_sortedSchools`, `_SchoolsPanel`, `_AdminSection.schools`	`_institutions`, `_openInstitution`, `_deleteInstitution`, `_sortedInstitutions`, `_InstitutionsPanel`, `_AdminSection.institutions`
`showSchoolDialog`, `_SchoolDialog`	`showInstitutionDialog`, `_InstitutionDialog`

- The URL did not change (`/api/institutions` already used that name), and the texts shown to the user were already "Instituciones".
- Flutter icon names such as `Icons.school_outlined` were intentionally **not** renamed.

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3. Testing

Check	Before the changes	After
`flutter analyze`	No issues	No issues ✅
`flutter test`	22 / 22	**28 / 28** ✅

The 2 tests that checked the old contract (`/api/auth/register` path, `PUT` answering 204) were updated, and 6 tests were added (`test/frontend_api_contract_test.dart`):

New test	Checks
Editing a question sends its images back	`imageUrl`, `justificationImageUrl` and `options[].imageUrl` are in the `PUT` body
Restoring the session needs a single refresh-token request	1 request to `/api/auth/refresh-token`, user returned, new tokens saved
Logout	Calls `/api/auth/logout` and clears local tokens
Revoking an invitation code	`PUT` with `isActive: false` that keeps `maxUses` and the expiration
Practice history	Uses `/api/practice-sessions/history` and parses the counts
Renamed routes	`/api/subscriptions/current-user` and `/api/reports/student-performance`

End-to-end check against the real backend (temporary test, not kept in the repo): the real repositories were run against the backend `refactor` branch — once with a fresh database and once with a copy of the developer's database:

- **Admin flow:** login, load the 5 lists, create / edit / delete subject, question (keeping images), institution and invitation code (including revoke), restore session, logout → ✅
- **Student flow:** login, subscription, plans, start practice, answer, history counts, finish, restore session, logout → ✅

A grep confirmed that no old route remains in `lib/` or `test/`.

After the `School` → `Institution` rename (2.6): `flutter analyze` no issues, `flutter test` 28 / 28, and the end-to-end check passed again against the backend with a fresh database and with a copy of the developer's database migrated at startup (the student now receives `institutionId` and `isInstitutional` is true).

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4. Pending

- The 2FA and Google sign-in endpoints exist in the backend but the app does not use them yet.
- `test/` names are still in Spanish (existing tests); new tests are in English.